SAFEAREA

SECURE PERSONAL DOCUMENT VAULT

Detailed Product Requirements Document

Product promise
Save, organize, protect, view and retrieve important personal documents from one calm, easy-to-use mobile vault.

Design direction
Follow the supplied SAFEAREA five-screen mockup and the reference app’s warm editorial visual language: cream surfaces, deep navy type, raspberry actions, soft green states, thin outlines and rounded cards.

Implementation intent
Give the AI coding agent clear instructions to reproduce the mockup faithfully, then implement real document capture, local storage, security controls and useful vault tools.

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Prepared as an implementation-ready product and UX specification.

0. Important implementation and security note

SAFEAREA handles highly sensitive identity information. A browser localStorage database is not a secure document vault. It is readable by scripts running in the same origin, has small storage limits, is not designed for document binaries, and does not provide encryption or biometric protection by itself.

For a prototype, localStorage may hold non-sensitive app preferences and document metadata only. Store document files in IndexedDB for a browser prototype, and encrypt files before persistence where feasible. For a production Android app, use app-private storage with encryption and Android Keystore; use Android BiometricPrompt for biometric confirmation. Never store PINs, passwords, biometric templates, raw identity scans or unencrypted document contents in localStorage.

The AI agent must not label the app “bank-grade”, “verified”, or “fully secure” unless those claims are technically implemented and independently reviewed. SAFEAREA is a personal storage tool; it does not authenticate government documents or verify their legal validity.

1. Product overview

SAFEAREA is a mobile-first personal document vault. Users can scan paper documents with the device camera, import existing files or photos, enter key details manually, organize documents, view saved records behind an access lock, and copy selected information when needed.

Primary users include people who need quick access to identity and everyday documents: driving licences, citizenship certificates, national ID cards, passports, education certificates, insurance papers, vehicle documents, address proofs, receipts and other personal files.

Product goals

• Make adding a document simple: scan, choose a file/photo, or enter details manually.

• Make retrieval fast: search, categories, favourites, recent documents and clear document details.

• Protect access with PIN, pattern and device biometric authentication where supported.

• Let users copy individual fields or a carefully selected set of fields without retyping.

• Offer useful built-in tools: scanner, crop/rotate, OCR-assisted field extraction, PDF preview, expiry reminders and secure export.

• Maintain a calm, consistent, accessible interface that closely matches the supplied mockup.

Non-goals

• Do not claim that a stored document is government-verified or legally equivalent to the original.

• Do not silently upload documents to a server or third-party OCR service.

• Do not collect more permissions than a feature needs.

• Do not implement fake biometric, fake scanning, fake OCR or simulated file operations.

2. Design reference and visual requirements

The AI agent must use the supplied SAFEAREA five-screen mockup as the primary visual source. Reproduce its composition, proportions, spacing, typography hierarchy, card outlines, button treatments and overall density before inventing additional screens. The supplied Nepali-learning UI reference informs the shared design language; do not turn SAFEAREA into a generic dark-gradient security dashboard.

• Base appearance: warm cream/beige canvas, deep navy text and outlines, raspberry primary actions, soft mint/green success states.

• Use flat, restrained surfaces. Avoid glossy 3D illustrations, excessive gradients, oversized empty hero art and decorative clutter.

• Use consistent rounded cards, pill-like actions, thin dark borders and simple outlined icons.

• Keep layouts mobile-first, compact and thumb-friendly. No desktop-first layout.

• Use the mockup's five-screen sequence as the initial product flow: Welcome, Security Setup, Home Vault, Document Details, Add Document.

• Use realistic sample placeholders only in preview/demo content. Never ship another person's real ID details.

Design tokens

Token	Default	Usage
Canvas	Warm cream #F1EADB	Main page background
Surface	Warm white #FFFDF7	Cards, sheets and document panels
Ink	Deep navy #171431	Headings, body text, borders
Primary	Raspberry #D5104B	Main CTA, active tab, key actions
Success	Soft green #438B68 / pale mint	Saved, unlocked, completed states
Accent	Muted yellow #F3B73E	Expiry / attention / secondary highlights
Border	Navy at low visual weight or warm grey	Card and control outlines
Typography	Readable sans-serif + Devanagari-capable fallback	Clear hierarchy and correct multilingual rendering

3. Theme system: pastel, light and dark

Provide a theme picker with at least the following coordinated pastel themes. Every theme must preserve contrast, semantic status colours and consistent component states.

Theme	Main canvas	Primary accent	Secondary accent
Cream (default)	#F1EADB	#D5104B raspberry	#78A88F sage
Mint	#EAF3EC	#347B62	#B7D9C4
Lavender	#F0ECF7	#7562A6	#C9BDE4
Powder Blue	#EAF1F7	#456F9B	#B9D2E8
Blush	#F8ECEB	#B94F68	#E9C1C8
Butter / Lemon	#F7F1DE	#A77A20	#E9D58B
Peach	#F8EDE4	#B86F4D	#EBC6AE
Sand	#EEE8DC	#806D55	#CFC1A9
Graphite dark	#171923	#E58AA4	#91BCA6
Midnight dark	#141629	#A9A2F0	#8CC7B0

• Include a visible Light / Dark / System segmented toggle in Settings.

• Theme selection and mode selection must persist across app restarts.

• Dark mode must be a complete designed theme, not a simple colour inversion. Maintain readable document previews and clear destructive-action contrast.

• Apply theme tokens consistently to navigation, forms, dialogs, scanner overlays, toasts, empty states and document viewer.

• Respect system appearance by default if the user chooses System.

4. Information architecture and navigation

Area	Purpose / contents
Home	Vault status, greeting, live date/time, recent documents, favourites, expiry reminders, quick Add action.
Documents	All documents, category filters, search, sort, tags, grid/list view.
Add	Scan with camera, import photo, import PDF/file, enter details manually.
Tools	Document scanner, crop/rotate, OCR review, PDF tools, QR/barcode reader where relevant, expiry calculator, secure export.
Profile / Settings	Security, theme, permissions, storage, backup/export, privacy, help, app lock settings.

Use a compact bottom navigation with four or five destinations. Keep the primary Add action prominent and reachable. Avoid duplicate destinations: Home is a summary, Documents is the complete library, Tools contains utilities, and Profile contains settings/security.

5. Screen-by-screen requirements

5.1 Welcome / onboarding

• SAFEAREA logo and concise value proposition: “Your personal document vault”.

• Use the shield/lock mark from the mockup, but keep illustration small and restrained.

• Primary action: Get Started. Secondary action: Learn how privacy works.

• Explain that documents stay on-device in the prototype unless the user explicitly exports or shares them.

• Do not ask for every permission on first launch. Request permissions contextually when the user starts the related feature.

5.2 Security setup / unlock

• Offer PIN, pattern and supported device biometrics (fingerprint / face authentication depending on OS).

• Allow users to choose one primary unlock method and an optional fallback PIN.

• Require PIN confirmation and provide a recovery warning before enabling app lock.

• Biometric unlock must use native platform APIs; do not imitate biometrics with a button or localStorage flag.

• Auto-lock on app background, screen timeout or configurable inactivity. Provide a clear manual Lock Now action.

• Rate-limit repeated incorrect PIN attempts and introduce increasing delays. Do not permanently lock users out without a recovery path.

5.3 Home vault

• Header: SAFEAREA brand, settings icon, and live local date/time.

• Vault status card: locked/unlocked state and selected protection method; never claim encryption unless enabled.

• My Documents section with See All and document count.

• Document rows/cards: icon, document title, issuing country/region or category, optional expiry date and last-updated label.

• Quick actions: Add Document, Scan, Search.

• Recent documents and favourites; hide sensitive field values in dashboard previews.

• Expiry reminder card for documents nearing expiry, with user-controlled notification settings.

5.4 Document details

• Show a preview of the front/back image or PDF, with tap-to-zoom and page navigation.

• Show structured fields in a readable list; support copy-one-field and Copy Selected / Copy All Info.

• Mask highly sensitive values by default (for example, show only the last few characters); require reveal confirmation.

• Actions: Edit Details, Replace File, Favourite, Share/Export, Move Category, Archive, Delete.

• Display document type, issue date, expiry date, created date, last modified date and optional notes.

• Clearly label OCR-extracted values as needing user review until confirmed.

5.5 Add document

• Three prominent paths: Scan Document, Choose from Gallery/Photos, Import File (PDF or supported image), plus Enter Manually.

• Show accepted file formats and maximum sizes based on actual platform limits.

• Allow front/back capture for two-sided IDs and multiple pages for certificates.

• After capture/import, show crop, rotate, perspective correction and retake controls.

• Ask the user to confirm extracted text before saving. Never silently accept OCR as authoritative.

5.6 Search, category and organisation

• Search by document title, category, tags and user-approved indexed fields.

• Filters: category, favourites, recently added, expiry status and file type.

• Categories should be editable; include Identity, Driving & Vehicle, Travel, Education, Finance, Health, Home & Address, Work and Other.

• Allow custom tags, rename, archive and duplicate detection prompts.

5.7 Profile and settings

• Security & Lock: enable/disable app lock, change PIN, reset pattern, biometric preference, auto-lock duration.

• Appearance: Light / Dark / System and pastel theme picker.

• Notifications: expiry reminders and optional security notifications.

• Storage: used space, document count, file type breakdown, clear cache (never delete original documents without confirmation).

• Privacy: data location, permission status, export data, delete all local data, privacy notice.

• About, support, app version and accessibility preferences.

6. Core feature requirements

Feature	Required behaviour
Native camera scanner	Open the real device camera; detect document edges if supported; capture, crop, rotate, perspective-correct and retake.
Gallery/photo import	Use OS photo picker or scoped media picker; support selecting multiple images where allowed.
File import	Use native document picker for PDF and supported image formats; preserve original file.
OCR	Extract text on-device where possible; present editable suggestions; allow manual correction; retain original scan.
Copy information	Copy individual fields or selected/all visible fields; show confirmation; offer a privacy-conscious clipboard clear timer where platform permits.
PDF preview	Render multi-page PDFs, zoom, page navigation and rotate view; never flatten or overwrite originals unexpectedly.
Expiry reminders	Store optional expiry date; schedule local notification with user-selected lead times.
Search and filters	Fast local search; filter/sort without network access.
Export and share	Explicit user action only; choose PDF/image or structured text export; show a warning before sharing sensitive content.
Backup / restore	For prototype, export an encrypted archive only if implemented correctly; do not imply cloud sync exists.

7. Camera, scanning and file handling

• Use the native camera stack for a native app (for Android, CameraX is the preferred implementation path). If the project is a web app, use supported browser camera APIs and clearly handle browser permission limitations.

• Scanner workflow: explain purpose → request camera permission → live camera preview → auto-edge detection (if available) → capture → crop/perspective correction → rotate → retake/confirm → OCR review → save.

• Support front and back sides, multi-page documents, flash/torch, camera switch only if useful, and a manual capture fallback.

• Preserve original capture and create optimized previews separately. Do not destructively overwrite the original.

• Validate file type and size, handle corrupt/unsupported files gracefully, and show upload/import progress.

• Do not transmit document files to an AI/OCR service unless the user explicitly opts in and the destination and privacy impact are disclosed.

8. Permissions and permission UX

Request permissions just in time, only when the user invokes a feature that needs them. Do not show a single “Allow everything” prompt or repeatedly pressure the user after denial.

Permission	When to ask	Fallback if denied
Camera	When user taps Scan Document.	Offer photo/file import or manual entry; show how to enable camera later.
Photos / gallery	When user chooses Import Photo.	Use system photo picker / selected-item access where available; do not demand broad gallery access if unnecessary.
Files / documents	When user chooses Import File or Export.	Use system document picker and scoped access. Avoid broad all-files permission unless strictly necessary and policy-compliant.
Notifications	When user enables expiry reminders or security alerts.	Keep reminders visible in-app; explain how to enable notifications later.
Biometrics	When user enables biometric unlock.	Offer PIN/pattern fallback; biometric enrollment remains controlled by device settings.

Important: “Full access to gallery/files” should not be requested by default. Prefer Android Photo Picker, Storage Access Framework/document picker and scoped access. Android's broad MANAGE_EXTERNAL_STORAGE permission is highly restricted and generally inappropriate for a private vault. Ask for only the access needed for the selected task.

9. Live date and time

• Display current device-local date and time on the home screen, updating while the app is active.

• Use the operating system clock and locale; do not hardcode a timestamp or rely on a server clock for a local display.

• Respect 12/24-hour preference and locale-specific date formatting.

• Update on app resume and when the device time zone or clock changes.

• Use a clear label such as “Today” or a formatted date; avoid displaying seconds unless there is a real product need.

10. Data model and localStorage requirements

The user requested localStorage as the backend database. Implement this only as a prototype/local-only persistence layer with the security boundary below. Do not represent it as a production-grade secure vault.

Entity	Suggested fields
AppSettings	themeId, appearanceMode, dateFormat, timeFormat, autoLockTimeout, notificationPreferences, onboardingComplete.
DocumentMetadata	id, title, category, issuer, country, tags, issueDate, expiryDate, createdAt, updatedAt, favourite, archived, notes, fileRef, mimeType, fileSize, pageCount.
DocumentField	id, documentId, label, value, isSensitive, isMasked, source (manual/OCR), userConfirmed.
DocumentAsset	id, documentId, blobKey, side/pageIndex, thumbnailKey, checksum, createdAt.
Reminder	id, documentId, reminderDate, enabled, notificationId, status.
AuditEvent (minimal)	id, eventType, timestamp; do not log raw document values or secrets.

Storage rules

• Use localStorage only for small, non-sensitive settings and metadata indexes in the prototype. Never put raw scans, PDFs, PINs, biometric data or unencrypted ID values in localStorage.

• Use IndexedDB for binary document files in a browser prototype. Add a documented storage abstraction so the prototype can later migrate to secure native storage.

• For a native Android build, use app-private files and encryption with keys protected by Android Keystore. Consider encrypted database storage for metadata if sensitive fields are persisted.

• Do not store a plain-text PIN or reversible PIN value. Use a vetted secure credential derivation design and platform-protected key material.

• Provide export, import, delete and clear-all operations with explicit confirmation. Deleting a document must remove its metadata, file, thumbnails and related reminders.

• Handle browser storage quota errors, unavailable storage, corrupted records and private browsing gracefully. Show a storage warning before data loss.

11. Security, privacy and trust

• Security is local-first and privacy by design. No analytics SDK should receive document names, field values, images or filenames.

• Lock the vault when backgrounded or after the configured idle timeout; hide sensitive app content in app switcher previews where the platform supports it.

• Mask sensitive fields by default and provide an explicit reveal control.

• Require re-authentication before exporting, sharing, deleting all data, changing lock settings or revealing especially sensitive values.

• Clipboard contents are sensitive: copy only on explicit tap, show a confirmation, and clear clipboard after a short interval where supported.

• Do not expose document previews or OCR text in push notification content.

• Provide a privacy notice that accurately describes local storage, backup, export and any network transmission.

• Perform threat modelling and security review before using SAFEAREA for real identity documents.

12. Built-in tools

Tool	Description / acceptance
Document Scanner	Camera capture, edge detection, crop, perspective correction, rotate, multi-page and retake.
OCR Text Extractor	On-device text recognition where available; editable result; confidence cues; user confirmation.
PDF Viewer	Multi-page preview, zoom, rotate, page count and safe export.
Image Tools	Crop, rotate, straighten, compress a copy and create thumbnails without changing original.
Field Copier	Copy one field, selected fields or all confirmed fields; preserve readable labels.
Expiry Tracker	Optional expiry dates, reminders, expired/expiring status and calendar-friendly date display.
QR / Barcode Reader	Optional scanner for codes on documents; never assume decoded content is authentic.
Secure Export	Export selected files or metadata with clear warnings and format choices.
Storage Manager	Show local storage use, file sizes, duplicate candidates and cache size.
Recovery / Data Export	Export a user-controlled backup; restore only after preview and confirmation.

13. Interaction and microcopy standards

• Use direct labels: “Scan document”, “Choose file”, “Copy all information”, “Lock vault”, “Show details”.

• Every destructive action must state what will be deleted and offer Cancel.

• Use visible loading, success, empty, permission-denied, offline, storage-full and error states.

• Never rely on colour alone: pair success/error states with icons and text.

• Use haptics only for meaningful actions and only when supported.

• Keep animations short and functional: card selection, scan progress, copy confirmation and unlock feedback.

• Make all controls reachable with one hand where practical; use adequate touch targets and dynamic text sizing.

14. Accessibility and localization

• Support screen readers with meaningful labels, logical focus order and accessible names for icon-only buttons.

• Maintain WCAG AA contrast targets for text and controls in every theme.

• Support system font scaling and avoid clipping in compact phone layouts.

• Use a Devanagari-capable font fallback and test Nepali conjuncts, matras and mixed Latin/Devanagari content.

• Format dates using locale and make ambiguous dates explicit (for example, show month names where needed).

• Do not hardcode Nepal-only document types; allow region-specific categories and custom document types.

15. Technical implementation guidance for the AI agent

First inspect the existing repository, framework, routing, styling system and available native capabilities. Do not replace the stack or invent integrations without checking the project. Implement in small, testable modules and preserve the supplied mockup's visual identity.

Layer	Implementation direction
UI	Component-based, responsive mobile-first screens matching the supplied mockup; shared design tokens and reusable cards/buttons/rows.
Navigation	Clear routes for onboarding, unlock, home, document list, details, scanner, import, tools and settings.
State	Centralized state for lock status, selected theme, document list, current document, filters and scanner workflow.
Persistence	Prototype: localStorage for preferences/non-sensitive metadata; IndexedDB for files. Native production: encrypted app-private storage.
Camera	Native camera API / Android CameraX for native Android; browser camera APIs only if web project.
Biometrics	Native BiometricPrompt / platform authentication. WebAuthn only where appropriate; do not fake device biometrics.
OCR	On-device OCR if available; adapter interface so OCR provider can be changed without rewriting UI.
Notifications	Local notifications for expiry reminders; permission requested when user enables reminders.
Testing	Unit tests for storage and date logic; UI tests for navigation, permissions, lock flows, import, scanner and copy actions.

AI agent implementation rules

• Treat the provided five-screen SAFEAREA mockup as the source of truth for visual composition. Match its screen proportions, spacing, typography, palette, outline weight, card radii and button hierarchy before extending the design.

• Do not substitute a generic dashboard template, banking UI, neon security aesthetic, glassmorphism or oversized gradient hero.

• Make every visible button functional. If a platform capability is unavailable, explain it and provide a real fallback instead of a fake interaction.

• Use realistic placeholder content only in development; label it clearly and never use real personal data.

• Keep all sensitive sample values fictional. Do not hardcode actual identity numbers or personal details.

• Build empty, loading, error, permission-denied and success states for each major flow.

• Do not claim localStorage provides encryption or biometric security. Separate app-lock UI from actual protected storage.

• Document any feature that cannot be implemented in the current stack and why.

16. Acceptance criteria / definition of done

ID	Acceptance criterion
UX-01	The five core screens visually match the supplied mockup's design language and mobile proportions.
UX-02	Light, dark and all pastel themes render consistently across every screen.
UX-03	Theme and appearance preferences persist after restart.
DOC-01	A user can add a document by camera scan, photo import, file import or manual entry.
DOC-02	A two-sided identity document and multi-page PDF can be saved and viewed.
DOC-03	Users can edit, rename, categorize, tag, favourite, archive and delete documents.
DOC-04	Document preview supports zoom and page navigation where relevant.
COPY-01	Users can copy one field, selected fields or all confirmed fields, with visible confirmation.
SEC-01	PIN/pattern setup and verification work; credentials are never stored as plain text.
SEC-02	Biometric unlock invokes the real platform biometric API and has a PIN fallback.
SEC-03	Vault auto-lock works after backgrounding or configured inactivity.
PERM-01	Camera, photos/files and notifications are requested contextually, with graceful denied-permission states.
TIME-01	Home displays current local date/time and updates on resume/time-zone changes.
TOOL-01	Scanner, OCR review, PDF preview, expiry tracker and export tools have working flows or are clearly marked as not yet supported.
DATA-01	Prototype persists settings and metadata locally; files are stored outside localStorage.
DATA-02	Storage-full, corrupted-data and import-failure cases are handled without silent data loss.
PRIV-01	No document content is transmitted externally without explicit consent.
A11Y-01	Screen-reader labels, scalable text and adequate contrast work in each theme.

17. Suggested delivery phases

Phase	Deliverables
1. Visual foundation	Design tokens, typography, navigation shell, five mockup screens, light/dark and pastel themes.
2. Vault essentials	Document list, add flows, metadata, file import, camera scan, preview, edit, search and categories.
3. Security	PIN/pattern, native biometric integration, auto-lock, masked fields, secure storage architecture.
4. Smart tools	OCR review, multi-page PDF, copy tools, expiry reminders, scanner enhancements and export.
5. Quality and release	Accessibility, localization, device testing, permission flows, backup/recovery, privacy and security review.

18. Final instruction to the AI coding agent

Build SAFEAREA as a polished, mobile-first personal document vault. Begin by reproducing the supplied five-screen UI mockup faithfully: cream background, navy typography and outlines, raspberry primary buttons, muted green success states, rounded cards, compact spacing and simple icons. Add a coordinated pastel theme library plus complete Light / Dark / System mode switching. Implement real camera scanning, photo/file import, document viewing, editable metadata, OCR review, copy actions, expiry reminders, search, categories and useful inbuilt tools. Show the device-local live date and time. Request camera, photo/file and notification permissions only when their related features are used, and provide clear fallbacks when denied. Use localStorage only for non-sensitive prototype settings and metadata; keep document files in IndexedDB for a web prototype or encrypted app-private storage for native production. Implement actual platform PIN/biometric security rather than simulated lock screens. Keep every feature functional, handle errors and empty states, and never claim that a document is verified or securely encrypted unless that capability is genuinely implemented.

End of document